

COMPARISON BETWEEN FLEXIBLE, ORGANIC AND PRINTED ELECTRONICS

Flexible	Organic	Printed
Large-area flexible electronics are based on the monolithic integration of electronic components on amorphous substrates, which typically results in products that are large in size with length scales ranging from a few centimetres (as in sensors) to a few decimetres (as in displays and lighting), to several metres (as in organic solar cell-based panels). This is in contrast to conventional microelectronics where integrated circuits are fabricated on crystalline substrates, and length scales are typically only a few millimetres.	Organic electronic devices have the potential to become a low-cost alternative to traditional inorganic electronic applications, due to low material utilisation and simple processing methods. To demonstrate technological maturity and competitiveness for these materials in the current market, hi-tech products are being manufactured using organic materials such as semiconductor, dielectric and conductive materials.	Printed electronics is a set of printing methods, including 3D printing, which are used to create electrical devices on various substrates. Printing typically uses common printing equipment suitable for defining patterns on material, such as screen printing, flexography, gravure, offset lithography and inkjet. By electronic industry standards, these are low-cost processes.

posable sensors or lab-on-a-chip for health monitoring, flexible displays and energy-efficient lighting, among others. Major application areas are consumer electronics, healthcare, military, energy, games, logistics and automotive products.

Flexible electronics have been cited in connection with an extraordinarily broad range of product applications where bendable and stretchable characteristics offer value.

Photovoltaics. Photovoltaic panels that conform to curved or otherwise irregular surfaces offer numerous advantages over heavy and rigid panels that account for most of today's photovoltaic electricity generation.

Lighting. OLEDs can be mounted on curved and bendable surfaces to provide lighting with versatility and energy efficiency superior to conventional forms of lighting. Although OLED lighting has not been widely introduced commercially, limited-release prototypes and commercial products have become available to demonstrate their potential and allow interested users to try out OLED technology.

Displays. Flexible displays will be developed that are unbreakable, waterproof, rugged and capable of being rolled up or folded for convenience. One much discussed potential application is the ability to transform smartphones into much better computing platforms, by adding a convenient larger display that can be rolled up or folded when not in use.

Another application is flexible, roll-to-roll produced e-paper, which has already enjoyed some success on a trial basis when used to produce shelf tags for supermarkets. That said, the flexible display market has not devel-

oped commercially as quickly as had been hoped, partially due to industry restructuring and competition from tablet computers.

Sensors. Sensors embedded in plastic tags are already being used in radio frequency identification (RFID) applications. Many RFID devices use antennae that are printed on flexible substrates. Sensors embedded in uniforms and other forms of clothing can be used for similar purposes.

Medical devices. Flexible electronics has the potential in many important applications, such as wearable health monitoring devices and medical implants. Medical patches and implants utilising bendable and stretchable sensors can potentially monitor a wide range of biological functions.

Smart textiles. Flexible electronics technology may enable the creation of fabrics that can alter their characteristics in response to external stimuli, whether mechanical, thermal, electrical, chemical or biological.

Defence applications. Flexible electronic devices printed onto uniforms, including electronic readers, displays featuring high-quality photographs, maps and other information, and communications devices, can drastically reduce the loads soldiers carry.

Flexible sensors can be placed on aircraft and vehicles to perform multiple functions, too.

Printed photovoltaic devices on tents, buildings and vehicles could generate local power, reducing the logistical demand for fuel.

Stretchable electronics. Electronic devices that can be stretched as well as bent without losing functionality have many potential applications. The

company Mc10 is reportedly commercialising stretchable skin developed at University of Illinois, Chicago. It contains sensors to monitor body functions and provide early warning of heart and brain problems before these occur, enabling real-time intervention by a doctor.

In the automotive sector, rigid electronic components can be fabricated by stretching these to form during plastic-moulding, yielding lighter, smaller and lower-cost parts, such as overhead lighting systems.

Flexible batteries. Highly-flexible batteries would enable the design of a broad range of products free from the constraints of rigidity and weight that characterise conventional batteries. Flexible batteries could power wearable electronics, including wristbands and clothing. These could enable the design of slimmer electronic products, and provide a power source for digital smart labels, such as freshness detectors on food packaging.

Growth drivers

Flexible electronic devices are easy to assemble, since these are lightweight and portable. This has resulted in the growing adoption of flexible electronic devices in areas such as solar power harnessing, healthcare and consumer electronics application, which are the key growth drivers.

There is significant demand across the globe for flexible and lighter products. Flexible electronic products that are in high demand include flexible displays, flexible photovoltaics, flexible batteries, flexible sensors, flexible memories and flexible RFID devices.

The main objective of manufacturers is to achieve large volumes and high-quality production processes